

International Islamic University Chittagong (IIUC)
Department of Computer and Communication Engineering
Midterm Examination

Program: **B.Sc (Engg.)**
 Course Code: **CCE-3501**
 Total Marks: **30**

Semester: **Spring-2026**
 Course Title: **Microprocessor and Assembly Language**
 Time: **1 Hours 30 Minutes**

(i) Answer all the questions. The figures in the right-hand margin indicate full marks.						
(ii) Course Learning Outcomes (COs) and Bloom's Levels are mentioned in additional Columns.						
Course Learning Outcomes (CLOs) of the Questions						
CLO 1	Gain knowledge of assembly level programming to write and test programs of moderate complexity for a real-time machine.					
CLO 2	Apply knowledge to design circuits for various applications using Microcontrollers.					
Bloom's Levels of the Questions						
Letter Symbols	R	U	Ap	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

- Q1** a) What is Microprocessor? Explain the working principle of CPU. CLO1 U 5
 b) Write the main differences between Von Neumann architecture and Harvard architecture and draw labelled block diagrams. CLO1 R 5
- Q2** a) Differentiate between a microprocessor, microcomputer, and microcontroller with examples. CLO1 R 5
 b) Explain the working principle of a multiuser system with example. Also discuss its advantages and disadvantages. CLO1 U 5
- Q3** a) Compare multiprogramming, multiprocessing, multiuser system, and parallel processing. CLO1 R 5
 b) Draw the architecture of 8085 and explain ALU, registers, and control unit. CLO2 Ap 5
- OR**
- Q3** a) Differentiate between 8085 and 8086 Microprocessor. CLO1 R 5
 b) Explain the workflow of the 8086 microprocessor and draw its labelled pin diagram. CLO2 Ap 5

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Semester: **Autumn-2025**
 Course Title: **Microprocessor and Assembly Language**
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Course Learning Outcomes (CLOs) of the Questions

- CLO1** Gain knowledge of assembly level programming to write and test programs of moderate complexity for a real-time machine.
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Bloom's Levels of the Questions

Letter Symbols
Meaning

R Remember **U** Understand **Ap** Apply **An** Analyze **E** Evaluate **C** Create

Q1.	a)	Describe the four essential operations performed by a CPU during instruction execution	CLO1	U	5
	b)	Elaborate on the architectural distinctions between the Von Neumann and Harvard architectures, with well-labeled block diagrams of both architectures.	CLO1	U	5
Q2.	a)	Differentiate between a microprocessor and a microcontroller in terms of architecture, functionality, and application areas. Support your answer with a comparison table and labeled diagrams of each.	CLO1	U	5
	b)	Discuss the concept of parallel processing in modern computer systems. How does it enhance computational speed and efficiency? Provide suitable examples of parallel architectures.	CLO2	Ap	5
Q3.	a)	Describe the main features of the Intel 8085 microprocessor.	CLO1	U	3
	b)	Design a simple data flow diagram that shows how an instruction is fetched and executed in an 8085 microprocessor.	CLO2	Ap	7
OR					
Q3.	a)	Compare the architectural differences between the 8085 and 8086 microprocessors.	CLO1	U	3
	b)	Design a simple data flow diagram that shows how an instruction is fetched and executed in an 8086 microprocessor.	CLO2	Ap	7

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Midterm Examination

Program: B.sc (Engg.)

Course Code: CCE-3501

Total Marks: 30

Semester: Spring-2025

Course Title: Microprocessor and Assembly Language

Time: 1 Hours 30 Minutes

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Course Learning Outcomes (CLOs) of the Questions	
CLO1	Gain knowledge of assembly level programming to write and test programs of moderate complexity for a real-time machine.
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		Bloom's Levels of the Questions						
Letter Symbols Meaning		R Remember	U Understand	Ap Apply	An Analyze	E Evaluate	C Create	
Q1.	a)	Explain the fundamental differences between the Von-Neumann and Harvard architectures with a labelled block diagram.				CLO1	U	5
	b)	How the evolution of microprocessors has impacted modern computing?				CLO1	U	2
	c)	Define the basic four working principle of CPU.				CLO1	R	3
Q2.	a)	Explain the differences between a microprocessor, microcomputer, and microcontroller with examples.				CLO1	U	5
	b)	Using the microcomputer block diagram, illustrate how input devices (e.g., keyboard) and output devices (e.g., monitor) interact with the CPU and memory in a simple data transfer task.				CLO2	Ap	5
Q3.	a)	What is Parallel Processing? How it works?				CLO1	U	4
	b)	Design a simple data flow diagram that shows how an instruction is fetched and executed in an 8085 microprocessor.				CLO2	Ap	6

OR

Q3.	a)	Compare 8085 with 8086 microprocessors.	CLO1	U	4
	b)	Design a simple data flow diagram that shows how an instruction is fetched and executed in an 8086 microprocessor.	CLO2	Ap	6

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Midterm Test

Program: **B.sc (Engg.)**

Course Code: **CCE-3501**

Total Marks: **15**

Semester: **Autumn-2024**

Course Title: **Microprocessor and Assembly Languages**

Time: **45 Minutes**

Q1.	Explain the signals SOD, Reset Out . Write an assembly language program to find the sum of 1 to A using a conditional jump.	CLO1	R,U	5
Q2.	Explain the instructions SHLD 8009 and INX H . Find the number of machine cycles, t-state and total execution time for the instruction MVI A,32H. MOV B,M;	CLO1	U	5
Q3.	Explain the instructions ADC E and LXI D . Write an assembly language program to interchange (swap) the contents of two memory locations 800A H and 800B H.	CLO1	U	5
	OR			
Q3.	How do the address and data lines are demultiplexed in 8085? Explain with proper diagram. If the 8085 adds A7H and 7DH, specify the contents of the accumulator and the status of the S, Z, and CY flags.	CLO1	U	5

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Midterm Examination

Program: B.sc (Engg.)
 Course Code: CCE-3501
 Total Marks: 30

Semester: Spring-2024
 Course Title: Microprocessor and Assembly Language
 Time: 1 Hour 30 Minutes

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Letter Symbols	R	U	Ap	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

Q1.	a)	What is a Microprocessor? Draw and explain the architecture of 8085.	CLO1	U	5
	b)	What is the difference between INR & INX instructions? How do the address and data lines are demultiplexed in 8085? Explain with proper diagram.	CLO1	U	5
Q2.	a)	Define Machine Cycle. Draw the Block diagram of DMA controller and explain its operations.	CLO1	U	5
	b)	Find the number of machine cycles, t-state and total execution time for the instruction MVI A,32H . Draw the timing diagram for any machine cycle of the instruction.	CLO1	Ap	5
Q3.	a)	Explain the signals HOLD , READY . Write an assembly language program to find the sum of 1 to 5 using conditional jump.	CLO1	Ap	5
	b)	Write the steps involved to fetch a byte in 8085. How many flags are there in flag register of 8085? Explain significance of each.	CLO1	Ap	5
	OR				
Q3.	a)	Explain the instructions LDAX and LXI . Write an assembly language program to interchange (swap) the contents of two memory locations 2100 H and 2101 H.	CLO1	Ap	5
	b)	State the control and status signals during different machine cycles. If the 8085 adds 87H and 79H, specify the contents of the accumulator and the status of the S, Z, and CY flags.	CLO1	Ap	5

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Letter Symbols Meaning		R Remember	U Understand	Ap Apply	An Analyze	E Evaluate	C Create		
Q1	a)	What is Microprocessor? Draw the pin diagram of 8085 μ P and explain the functionality of following pins: Reset out, Ready, ALE				CLO1	R	1+5	
	b)	Define instruction cycle. State the control and status signals during different machine cycles.				CLO1	R	4	
Q2	a)	What is a bus? Draw the architecture of 8085 μ P				CLO1	R	5	
	b)	Draw the timing diagram of Memory Read. Explain demultiplexing of address and data bus with appropriate figure.				CLO1	Un	2+3	
Q3	a)	Define program counter. State the flags of the flag register for the following instructions. MVI A, XH ; X= last 2 digits of your ID MVI B, EFH; ADD B;				CLO1	R/ Un	1+3	
	b)	Make a short note on: i. Opcode Fetch ii. T State Explain the concept of DMA with appropriate figure				CLO1	R	3+3	
OR									
Q3	a)	Write a program to subtract 2 numbers and store the result and value of carry flag in different memory locations. Assume that the last number is the last two digits of your ID.				CLO1	Un	4	
	b)	Explain the following instructions with examples LDA, ADC Find the number of machine cycles, t-state and total execution time for the following instructions and explain them in detail. a) MOV A, M b) MVI B, 25H				CLO1	Un/ Ap	2+4	

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Bloom's Levels of the Questions

	Letter Symbols Meaning	Bloom's Levels of the Questions								
		R Remember	U Understand	Ap Apply	An Analyze	E Evaluate	C Create			
1) a)	Differentiate between microprocessor, microcomputer, and microcontroller. Write the functions of the following pins: i) ALE ii) Reset Out iii) INTA					CLO1 CLO2	R	2+3		
1) b)	What is the difference between ADD and ADC instructions? Explain with examples. Make a short note on: i) Machine Cycle ii) T-state					CLO1 CLO2	R/UN	2+3		
2) a)	Explain demultiplexing of address and data bus with appropriate figure. Draw the timing diagram of Opcode fetch.					CLO1 CLO2	U/R	3+2		
2) b)	For each of the following instructions, give the new destination contents and the new settings of CF, SF, ZF, PF and OF. Suppose that flags are initially 0 in each part of this question. I. ADD AL, BL ; AL=0D H and BL=EF H II. SUB AL, BL; AL=9B H and BL=1A H					CLO1	Ap	5		
3) a)	Explain the concept of DMA with appropriate figure. State the control and status signals during different machine cycles.					CLO1 CLO2	U	3+2		
3) b)	Write the steps involved to fetch a byte in 8085. Explain the following instructions with examples i) LDAX ii) INX					CLO1	U/Ap	3+2		
OR										
3) a)	Write a program to add two 8-bit numbers and store the result and value of carry flag in different memory location. Assume that the last number is the last two digits of your ID. Write a program that has a finite loop with 14 iterations.					CLO1	U/Ap	3+2		
3) b)	Define instruction cycle. Find the number of machine cycles, t-state and total execution time for the following instructions • MVI A, 05H • MOV A, B					CLO1 CLO2	U/Ap	5		